EXERCISE (O)

SINGLE CORRECT TYPE QUESTIONS

Basics of Semiconductor

1. The manifestation of band structure in solids is due to -

(A) Bohr's correspondence principle. (B) Pauli's exclusion principle.

(C) Heisenberg's uncertainty principle. (D) Boltzmann's law.

PSD028

2. If the ratio of the concentration of electrons to that of holes in a semiconductor is $\frac{7}{5}$ and the ratio of currents is $\frac{7}{4}$ then what is the ratio of their drift velocities -

(A) $\frac{5}{4}$ (B) $\frac{4}{7}$ (C) $\frac{5}{6}$ (D) $\frac{4}{5}$

PSD032

3. Mobility of electrons in a semiconductor is defined as the ratio of their drift velocity to the applied electric field. If, for an *n*-type semiconductor, the density of electrons is $10^{19} m^{-3}$ and their mobility is 1.6 $m^2/(V.s)$ then the resistivity of the semiconductor (since it is an *n*-type semiconductor contribution of holes is ignored) is close to:

(A) $2\Omega m$ (B) $0.4\Omega m$ (C) $4\Omega m$ (D) $0.2\Omega m$

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PSD035

4. For extrinsic semiconductors, when doping level is increased:

(A) Fermi-level of *p*-type semiconductor will go upward, and Fermi-level of *n*-type semiconductors will go downward.

(B) Fermi-level of *p*-type semiconductors will go downward, and Fermi-level of *n*-type semiconductor will go upward.

(C) Fermi-level of both *p*-type and *n*-type semiconductors will go upward for $T > T_F$ *K* and downward for $T < T_F$ *K*, where T_F is Fermi temperature.

(D) Fermi-level of *p* and *n*-type semiconductors will not be affected.

PSD036

PN Junction Diode

5. In a *P*-*N* junction diode not connected to any circuit -

(A) the potential in the same everywhere.

(B) the *P*-type side is at a higher potential then the *N*-type side.

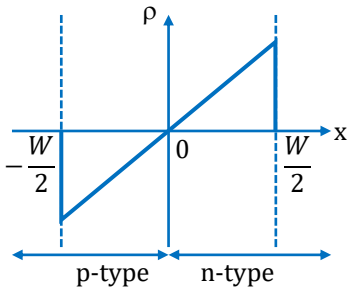
(C) there is an electric field at the junction directed from the *N*-type side to the *P*-type side.

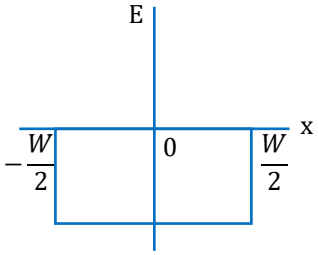
(D) there is an electric field at the junction directed from the *P*-type to the *N*-type side.

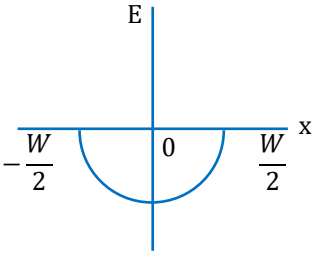
PSD030

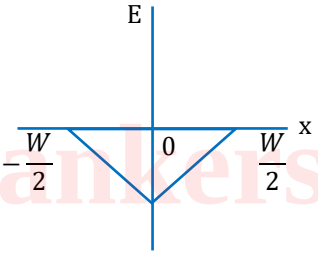
For More Join: @IITJEE_Advanced

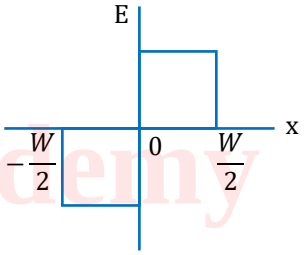
6. A semiconductor *pn* junction at thermal equilibrium has the space charge density ρ(x) profile as shown in the figure. The figure that best depicts the variation of the electric field E with x is :- (W denotes the width of the depletion layer)



(A) 

(B) 

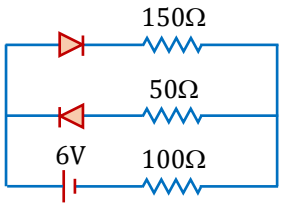
(C) 

(D) 

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PSD096

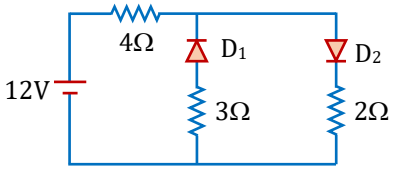
7. In the below given arrangement determine the ammeter reading, if each diodes have a forward resistance of 50Ω and infinite backward resistance –



(A) zero (B) 0.02 (C) 0.03 (D) 0.036

PSD040

8. The circuit has two oppositely connected ideal diodes in parallel. What is the current flowing in the circuit ?

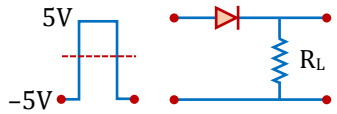


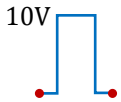
(A) 2.31 *A* (B) 1.33 *A* (C) 1.71 *A* (D) 2.00 *A*

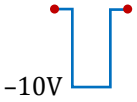
PSD047

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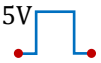
9. If in a *p-n* junction diode, a square input signal of 10*V* is applied as shown. Then the output signal across R_L will be :-



(A) 

(B) 

(C) 

(D) 

PSD049

Zener Diode

10. Most important use of Zener diode is to have –

(A) constant voltage across applied load.

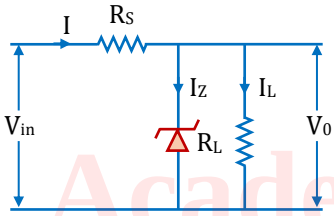
(B) any desired current at constant voltage.

(C) a *p-n* junction working under constant regulated voltage conditions.

(D) a *p-n* junction to operate at high voltages.

PSD044

11. In given figure when input voltage increases,



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(A) the current through R_S, R_L and Zener increases.

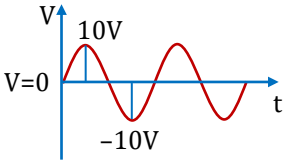
(B) the current through R_S increases, Zener increases but through R_L remains constant.

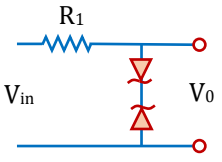
(C) the current through R_S increases, through Zener decreases, R_L increases.

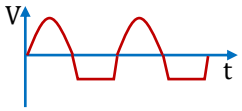
(D) the current through R_S increases, through Zener remains constant but R_L increases.

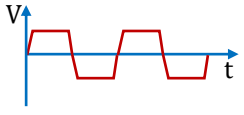
PSD048

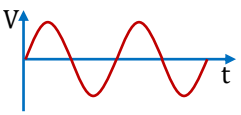
12. Take the breakdown voltage of the Zener diode used in the given circuit as 6*V*. For the input voltage shown in figure below, the time variation of the output voltage is : (Graphs drawn are schematic and not to scale)

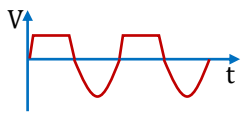




(A) 

(B) 

(C) 

(D) 

PSD053

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13. Zener breakdown occurs in a *p-n* junction having *p* and *n* both:

(A) lightly doped and have wide depletion layer.

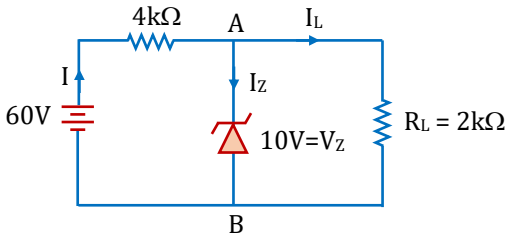
(B) heavily doped and have narrow depletion layer.

(C) lightly doped and have narrow depletion layer.

(D) heavily doped and have wide depletion layer.

PSD054

14. A Zener diode is connected to a battery and a load as shown below. The currents I, I_Z and I_L are respectively.



(A) 12.5 *mA*, 5 *mA*, 7.5 *mA*

(B) 15 *mA*, 7.5 *mA*, 7.5 *mA*

(C) 12.5 *mA*, 7.5 *mA*, 5 *mA*

(D) 15 *mA*, 5 *mA*, 10 *mA*

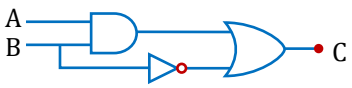
PSD055

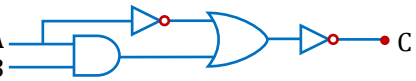
Logic Gates

15. Which of the following circuits correctly represents the following truth table?

A	B	C
0	0	0
0	1	0
1	0	1
1	1	0

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(A) 

(B) 

(C) 

(D) 

PSD085

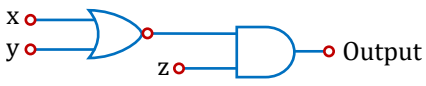
16. This symbol represents



(A) NOT gate (B) OR (C) AND gate (D) NOR gate

PSD087

17. In which of the following cases, we would obtain an output of one.

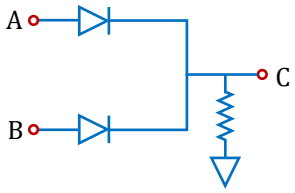


(A) $x = 1, y = 1, z = 1$ (B) $x = 1, y = 1, z = 0$ (C) $x = 0, y = 1, z = 1$ (D) $x = 0, y = 0, z = 1$

PSD089

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18. In the circuit below, A and B represent two inputs and C represents the output.

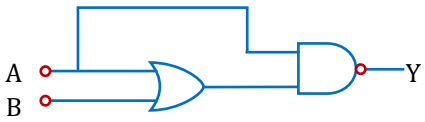


The circuit represents.

(A) AND gate (B) NAND gate (C) OR gate (D) NOR gate

PSD090

19. The truth table for the circuit given in the fig. is:



(A)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(B)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(C)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(D)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

PSD097

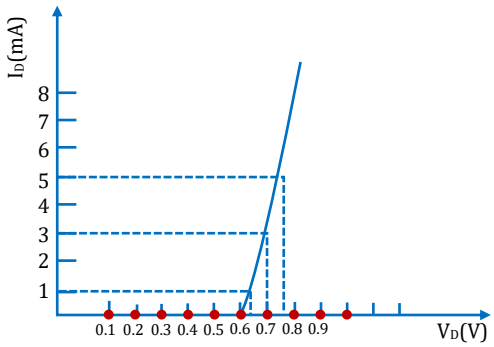
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EXERCISE (S)

PN Junction Diode

1. For the forward biased diode characteristics shown in the figure, the dynamic resistance at $I_D = 3\ mA$ will be_____Ω.

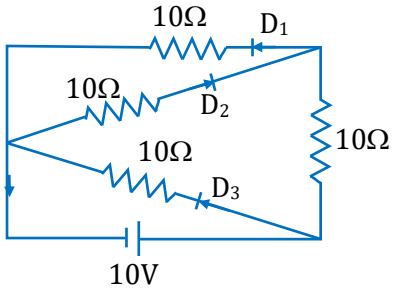


PSD114

2. When a diode is forward biased, it has a voltage drop of 0.5 *V*. The safe limit of current through the diode is 10 *mA*. If a battery of emf 1.5 *V* is used in the circuit, the value of minimum resistance to be connected in series with the diode so that the current does not exceed the safe limit is ___Ω.

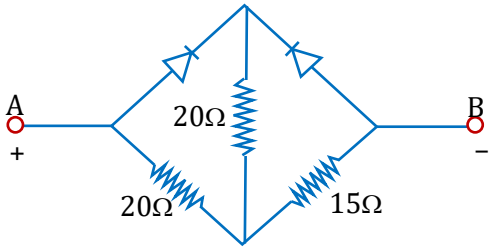
PSD123

3. In the given circuit, the current (I) through the battery will is $m\times10^{-1}\ A$ .Then the value of *m* will be;



PSD108

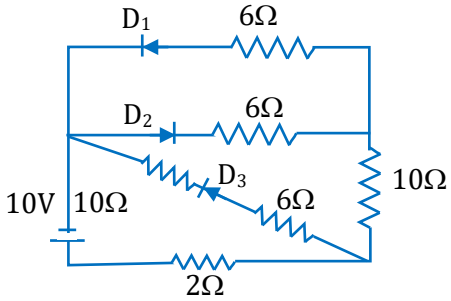
4. Two ideal diodes are connected in the network as shown in figure. The equivalent resistance between A and B is _____Ω.



PSD109

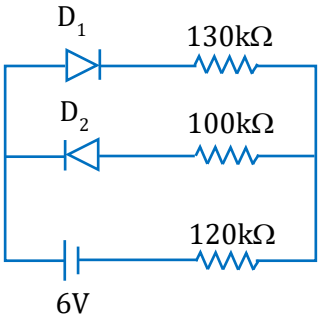
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5. As per the given circuit, the value of current through the battery will be ..... A.



PSD110

6. The circuit contains two diodes each with a forward resistance of 50 Ω and with infinite reverse resistance. If the battery voltage is 6 *V*, the current through the 120 Ω resistance is_ *mA*.



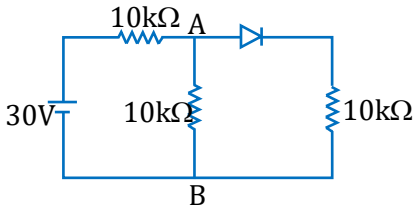
PSD113

7. Consider a situation in which reverse biased current of a particular *P-N* junction increases when it is exposed to a light of wavelength ≤ 621 *nm*. During this process, enhancement in carrier concentration takes place due to generation of hole-electron pairs. The value of band gap is nearly (in *eV*).

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PSD115

8. In the figure, potential difference between *A* and *B* is (in Volt):



PSD122

9. If a semiconductor photodiode can detect a photon with a maximum wavelength of 400 *nm*, then its band gap energy is nearly (*eV*) :

Planck's constant $h = 6.63 \times 10^{-34}$ J.s.

Speed of light $c = 3 \times 10^{8}\ m/s$

PSD124

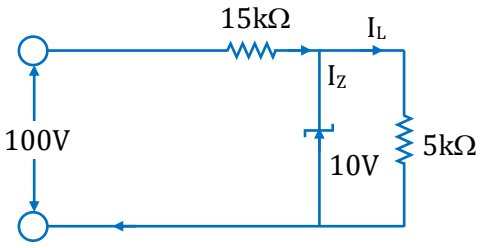
10. A light emitting diode (LED) is fabricated using GaAs semiconducting material whose band gap is 1.42 eV. The wavelength (in nm) of light emitted from the LED is nearly:

PSD107

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Zener Diode

11. For the circuit shown in figure, find:



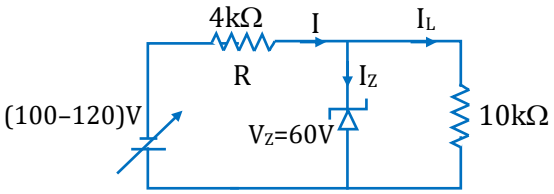
(i) the output voltage

(ii) the voltage drop across series resistance

(iii) the current through Zener diode.

PSD098

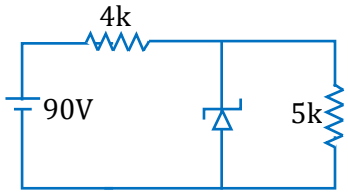
12. In the circuit shown below, maximum Zener diode current will be ___*mA*



PSD112

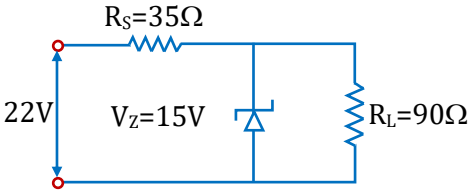
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13. The Zener diode has a V_z = 30 *V*. The current passing through the diode for the following circuit is ..... *mA*.



PSD116

14. The value of power dissipated across the Zener diode (V_z = 15 *V*) connected in the circuit as shown in the figure is $x \times 10^{-1}$ watt.

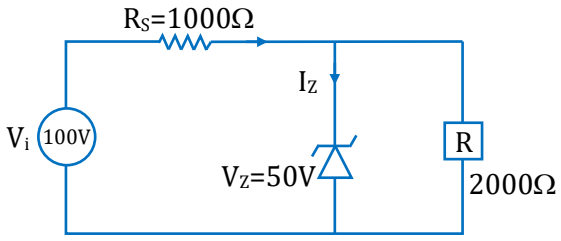


The value of *x*, to the nearest integer, is _____.

PSD117

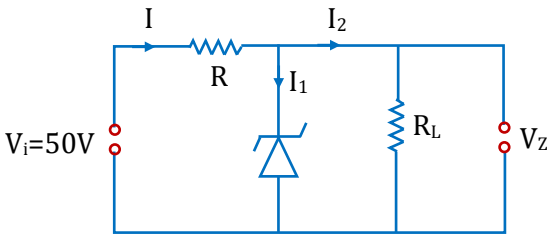
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15. For the circuit shown below, calculate the value of I_z (in mA) :



PSD118

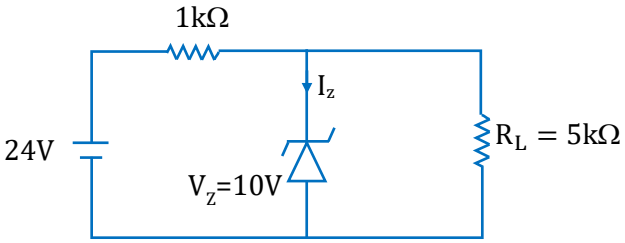
16. In a given circuit diagram, a 5 V Zener diode along with a series resistance is connected across a 50 V power supply. The minimum value of the resistance required, if the maximum Zener current is 90 mA will be ____ Ω.



PSD119

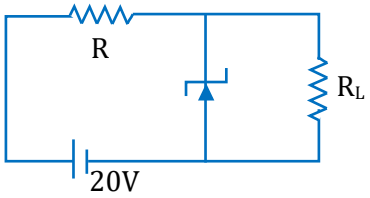
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17. For the given circuit, the power across Zener diode is ..... mW.



PSD120

18. A 8V Zener diode along with a series resistance R is connected across a 20 V supply (as shown in the figure). If the maximum Zener current is 25 mA, then the minimum value of R will be ____Ω.

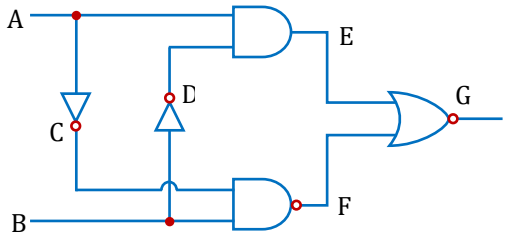


PSD111

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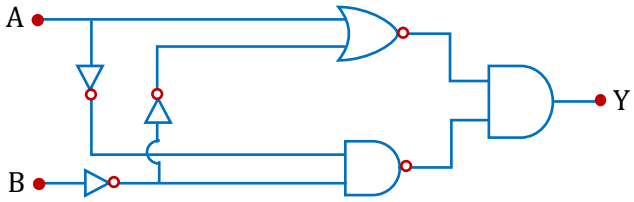
Logic Gates

19. Write a truth table for the circuit in figure, including the states at C, D, E, F and G.



PSD099

20. In the logic circuit shown in the figure, if input *A* and *B* are 0 to 1 respectively, the output at *Y* would be '*x*'. The value of *x* is _____.



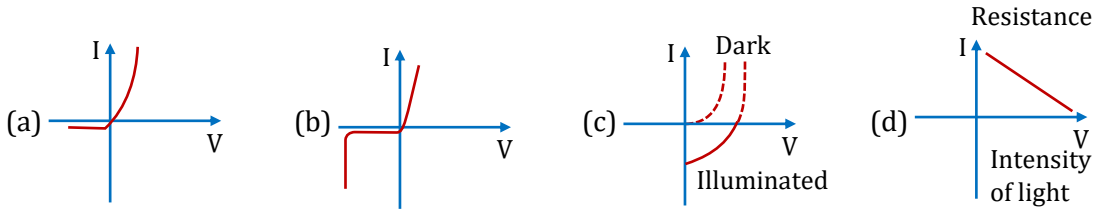
PSD121

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For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Identify the semiconductor devices whose characteristics are given below, in the order (*a*), (*b*), (*c*), (*d*): **[JEE (Main) 2016]**



(1) Zener diode, Solar cell, Simple diode, Light dependent resistance

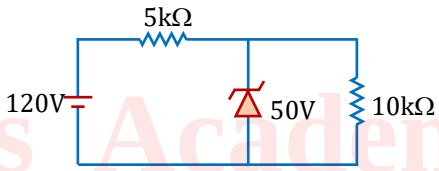
(2) Simple diode, Zener diode, Solar cell, Light dependent resistance

(3) Zener diode, Simple diode, Light dependent resistance, Solar cell

(4) Solar cell, Light dependent resistance, Zener diode, Simple diode

PSD021

2. For the circuit shown below, the current through the Zener diode is: **[JEE (Main) 2019]**



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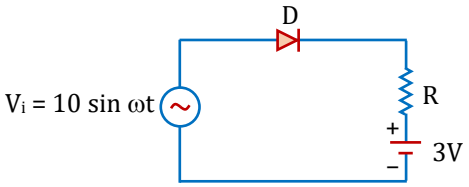
(1) 5 *mA* (2) Zero (3) 14 *mA* (4) 9 *mA*

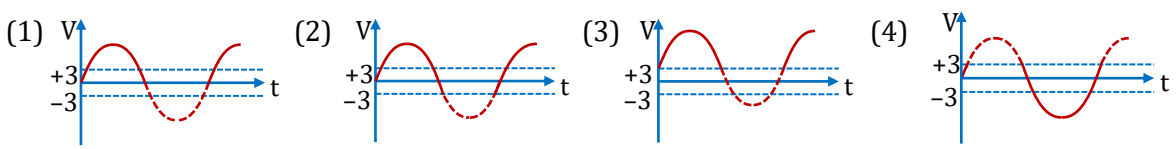
PSD022

3. In a semiconductor, the number density of intrinsic charge carriers at 27°*C* is $1.5 \times 10^{16}/m^3$. If the semiconductor is doped with impurity atom, the hole density increases to $4.5 \times 10^{22}/m^3$. The electron density in the doped semiconductor is _____ $\times 10^9/m^3$. **[JEE (Main) 2021]**

PSD023

4. Choose the correct waveform that can represent the voltage across *R* of the following circuit, assuming the diode is ideal one: **[JEE (Main) 2021]**





PSD024

For More Join: @IITJEE_Advanced

5. **Statement-I:**

To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect a capacitor across the output parallel to the load R_L.

Statement-II :

To get a steady *dc* output from the pulsating voltage received from a full wave rectifier we can connect an inductor in series with R_L.

In the light of the above statements, choose the most appropriate answer from the options given below: **[JEE (Main) 2021]**

(1) Statement I is true, but Statement II is false.

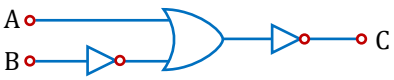
(2) Statement I is false, but Statement II is true.

(3) Both Statement I and Statement II are false.

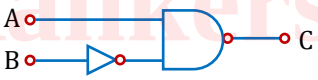
(4) Both Statement I and Statement II are true.

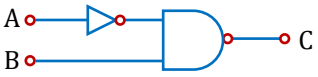
PSD025

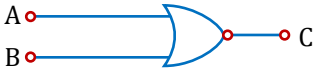
6. The logic circuit shown above is equivalent to: **[JEE (Main) 2021]**

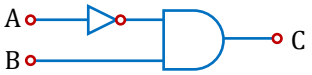


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(1) 

(2) 

(3) 

(4) 

PSD026

7. A potential barrier of 0.4 V exists across a p-n junction. An electron enters the junction from the n-side with a speed of $6.0 \times 10^5\ ms^{-1}$. The speed with which electron enters the p side will be $\frac{x}{3} \times 10^5\ ms^{-1}$ the value of x is _____.

(Given mass of electron = $9 \times 10^{-31}\ kg$, charge on electron = $1.6 \times 10^{-19} C$.)

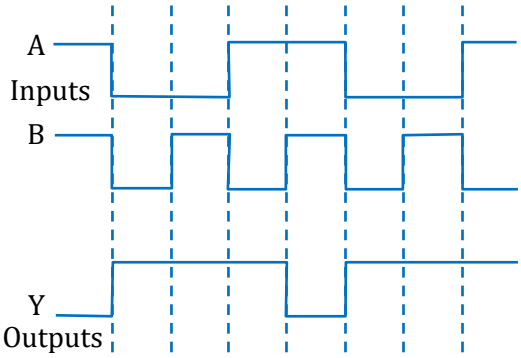
[JEE (Main) 2022]

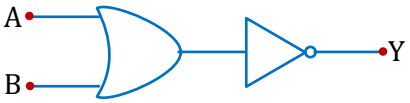
PSD125

For More Join: @IITJEE_Advanced

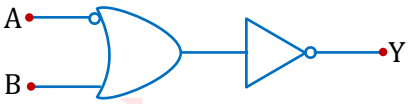
8. Identify the correct Logic Gate for the following output (Y) of two inputs A and B.

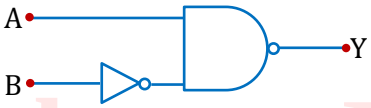
[JEE (Main) 2022]



(1) 

(2) 

(3) 

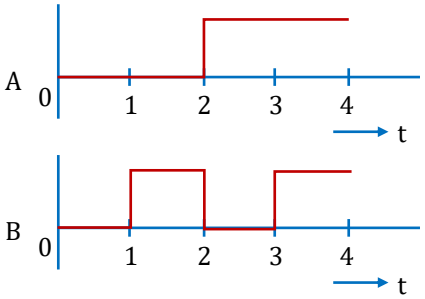
(4) 

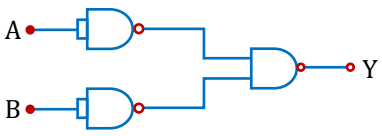
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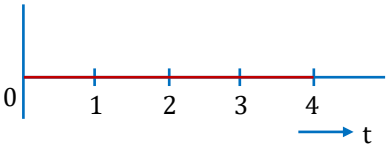
PSD126

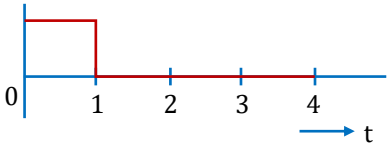
9. For the logic circuit shown, the output waveform at Y is:

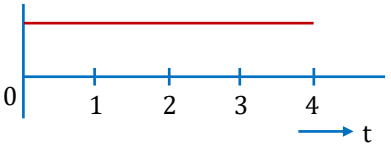
[JEE (Main) 2023]

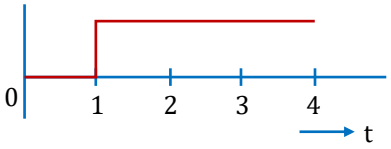




(1) 

(2) 

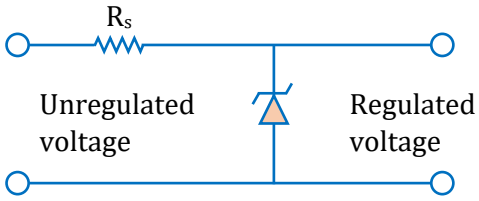
(3) 

(4) 

PSD100

For More Join: @IITJEE_Advanced

10. A zener diode of power rating $1.6W$ is to be used as voltage regulator. If the zener diode has a breakdown of $8V$ and it has to regulate voltage fluctuating between $3V$ and $10V$. The value of resistance R_s for safe operation of diode will be : **[JEE (Main) 2023]**



(1) 13.3Ω (2) 12Ω (3) 10Ω (4) 13Ω

PSD101

11. **Statement I :** When a Si sample is doped with Boron, it becomes P-type and when doped by Arsenic it becomes N-type semi conductor such that P-type has excess holes and N-type has excess electrons.

Statement II : When such P-type and N-type semi-conductors, are fused to make a junction, a current will automatically flow which can be detected with an externally connected ammeter.

In the light of above statements, choose the most appropriate answer from the options given below. **[JEE (Main) 2023]**

Options:

(1) Both Statement I and statement II are incorrect

(2) Statement I is incorrect but statement II is correct

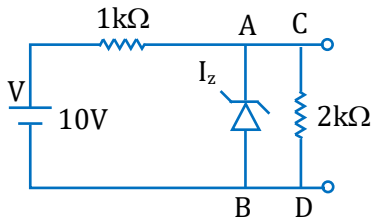
(3) Both Statement I and statement II are correct

(4) Statement I is correct but statement II is incorrect

PSD127

Rankers Academy JEE

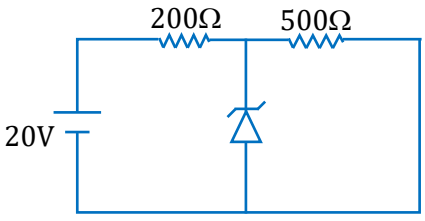
12. In the given circuit, the breakdown voltage of the Zener diode is 3.0 V. What is the value of I_z? **[JEE (Main) 2024]**



(1) 3.3 mA (2) 5.5 mA (3) 10 mA (4) 7 mA

PSD128

13. A Zener diode of breakdown voltage $10V$ is used as a voltage regulator as shown in the figure. The current through the Zener diode is **[JEE (Main) 2024]**



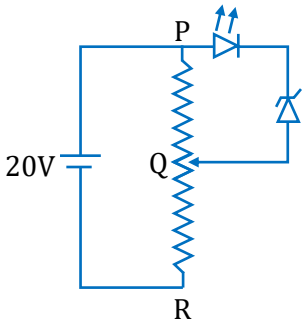
(1) 50 mA (2) 0 (3) 30 mA (4) 20 mA

PSD129

For More Join: @IITJEE_Advanced

14. A potential divider circuit is connected with a dc source of 20 *V*, a light emitting diode of glow in voltage 1.8 *V* and a zener diode of breakdown voltage of 3.2 *V*. The length (*PR*) of the resistive wire is 20 *cm*. The minimum length of *PQ* to just glow the LED is ..... cm.

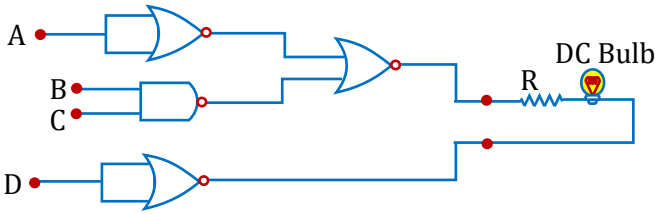
[JEE (Main) 2024]



PSD130

15. Following gates section is connected in a complete suitable circuit.

[JEE (Main) 2024]



For which of the following combination, bulb will glow (ON):

(1) $A = 0, B = 1, C = 1, D = 1$ (2) $A = 1, B = 0, C = 0, D = 0$

(3) $A = 0, B = 0, C = 0, D = 1$ (4) $A = 1, B = 1, C = 1, D = 0$

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PSD131

16. Conductivity of a photodiode starts changing only if the wavelength of incident light is less than 660 *nm*. The band gap of photodiode is found to be $\left(\frac{X}{8}\right)eV$. The value of *X* is :

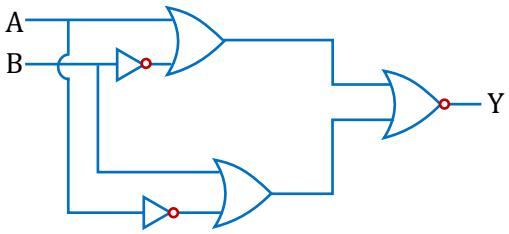
(Given, $h = 6.6 \times 10^{-34}\ Js$, $e = 1.6 \times 10^{-19} C$)

[JEE (Main) 2024]

(1) 15 (2) 11 (3) 13 (4) 21

PSD132

17. The output of the given circuit diagram is **[JEE (Main) 2024]**



(1)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

(2)

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	0

(3)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	0

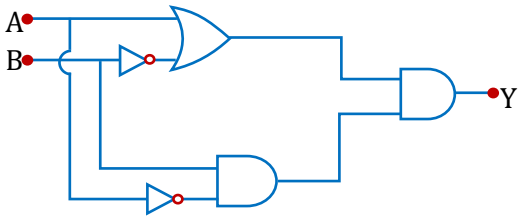
(4)

A	B	Y
0	0	0
1	0	0
0	1	1
1	1	0

PSD133

For More Join: @IITJEE_Advanced

18. The output *Y* of following circuit for given inputs is :



[JEE (Main) 2024]

(1) A•B(A + B) (2) A • B (3) 0 (4) $\bar{A}$•B

PSD134

19. Consider the following statements :

A. The junction area of solar cell is made very narrow compared to a photo diode.

B. Solar cells are not connected with any external bias.

C. LED is made of lightly doped p-n junction.

D. Increase of forward current results in continuous increase of LED light intensity.

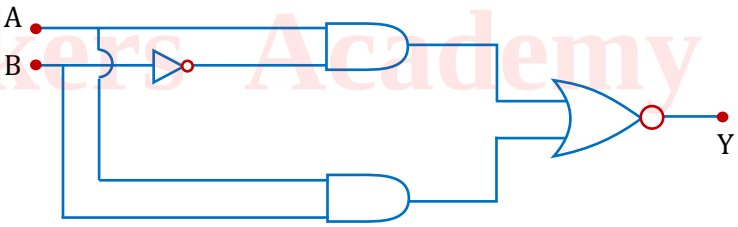
E. LEDs have to be connected in forward bias for emission of light.

[JEE (Main) 2025]

(1) B, D, E Only (2) A, C Only (3) A, C, E Only (4) B, E Only

PSD155

20. Which of the following circuits has the same output as that of the given circuit?



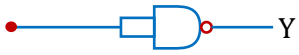
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[JEE (Main) 2025]

(1) 

(2) 

(3) 

(4) 

PSD156

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	B	B	C	B	B	D	D	A
Que.	11	12	13	14	15	16	17	18	19	
Ans.	B	B	B	C	B	A	D	C	A	

EXERCISE (S)

1. 25 **2.** 100 **3.** 15 **4.** 25

5. 1 **6.** 20 **7.** 2 **8.** 10

9. 3eV **10.** 875nm **11.** (i) 10 V (ii) 90 V (iii) 4 mA

12. 9 **13.** 9 **14.** 5 **15.** 25 mA

16. 500 **17.** 120 **18.** 480

19.

A	B	C	D	E	F	G
0	0	1	1	0	0	1
0	1	1	0	0	1	0
1	0	0	1	1	0	0
1	1	0	0	0	0	1

20. 0

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	5	3	4	4	14	2	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	3	5	2	1	3	3	4	1

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Important Notes

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